Contracted Farey expansions

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Regular CFs

Each $x \in \mathbb{R} \setminus \mathbb{Q}$ has a unique **regular continued fraction** (RCF) expansion

$$x = [a_0; a_1, a_2, \dots] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \ddots}},$$

where $a_n \in \mathbb{Z}$ and $a_n > 0$ for $n > 0$, understood as the limit of RCF-**convergents** $\frac{p_n}{q_n} = [a_0; a_1, \dots, a_n] \in \mathbb{Q}$. These satisfy

$$p_{n+1} = a_{n+1} p_n + p_{n-1},$$
$$q_{n+1} = a_{n+1} q_n + q_{n-1},$$

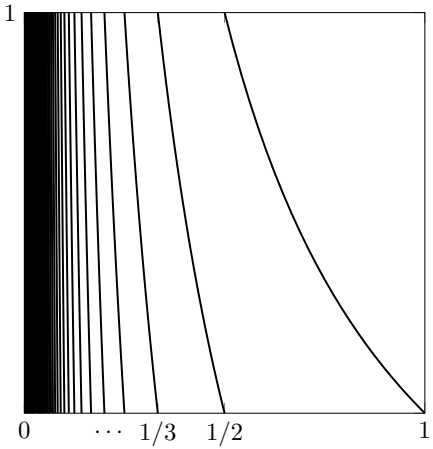
where $\left(\begin{smallmatrix} p_{-1} & p_0 \\ q_{-1} & q_0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & a_0 \\ 0 & 1 \end{smallmatrix}\right)$.

The Gauss map

The **Gauss map** $G : [0,1] \to [0,1]$ is defined by $G(0) = 0$ and for $x \neq 0$,

$$G(x) = \begin{pmatrix} 0 & 1 \\ 1 & a(x) \end{pmatrix}^{-1} \cdot x = \frac{1}{x} - a(x),$$

where $a(x) = \lfloor 1/x \rfloor$. Rearranging,



$$x = \frac{1}{a(x) + G(x)} = \frac{1}{a(x) + \dfrac{1}{a(G(x)) + G^2(x)}} = \dots,$$

i.e., $a_n = a(G^{n-1}(x))$ for $n > 0$. Symbolically,

$$G([0; a_1, a_2, \dots]) = [0; a_2, a_3, \dots].$$

The Gauss measure

Statistics of RCF-expansions are obtained with the G-invariant, a.c., ergodic probability measure ν_G, defined on Borel sets $A \subset [0,1]$ by

$$\nu_G(A) := \int_A \frac{dx}{\log 2(1+x)}.$$

For instance, Birkhoff's ergodic theorem implies

$$\lim_{n\to\infty} \frac{1}{n} \#\{1 \le k \le n \mid a_k = 1\} = \nu_G((1/2,1]) = 0.414037\ldots \quad \text{a.s.}$$

This measure was mysteriously mentioned by Gauss in an 1812 letter to Laplace. Since then, several authors have 'derived' ν_G.

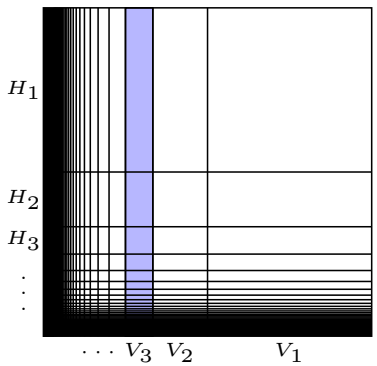
The natural extension of the Gauss map

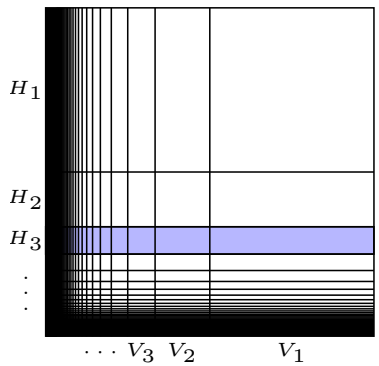
In the 1970s and 80s, Nakada, Ito and Tanaka introduced an explicit **natural extension** $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ of $([0,1], \mathcal{B}, \nu_G, G)$, i.e., the 'smallest' invertible dynamical system of which $([0,1], \mathcal{B}, \nu_G, G)$ is a factor. Here $\Omega = [0,1]^2$,

$$\mathcal{G}(x,y) = \left(\left(\begin{smallmatrix} 0 & 1 \\ 1 & a(x) \end{smallmatrix}\right)^{-1} \cdot x, \left(\begin{smallmatrix} 0 & 1 \\ 1 & a(x) \end{smallmatrix}\right) \cdot y \right), \quad \bar{\nu}_G(B) = \iint_B \frac{dxdy}{\log 2 (1+xy)^2}.$$

One has $\nu_G = \pi_* \bar{\nu}_G$, where $\pi(x,y) = x$. Symbolically,

$$\mathcal{G}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = ([0; a_2, a_3, \dots], [0; a_1, b_1, b_2, \dots]).$$





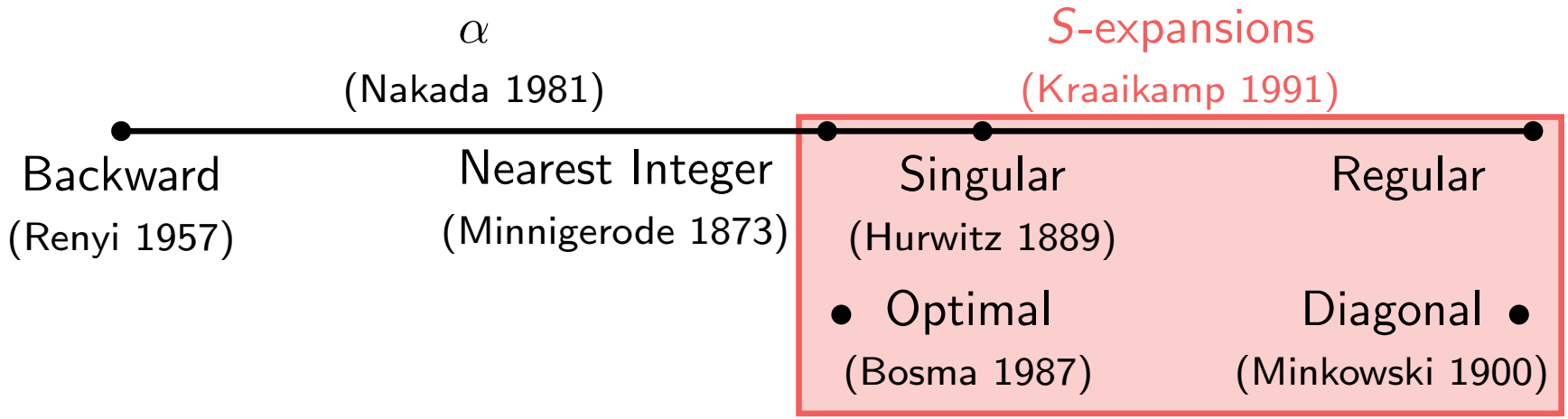
$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{G}} & \Omega \\ \pi \downarrow & & \downarrow \pi \\ [0,1] & \xrightarrow{G} & [0,1] \end{array}$$

Other CF-algorithms

Several other algorithms produce **generalised** CF (GCF) expansions

$$x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots] = \beta_0/\alpha_0 + \cfrac{\alpha_1/\alpha_0}{\beta_1 + \cfrac{\alpha_2}{\beta_2 + \ddots}}$$

with $\alpha_n, \beta_n \in \mathbb{Z}$, $\alpha_n \neq 0$.

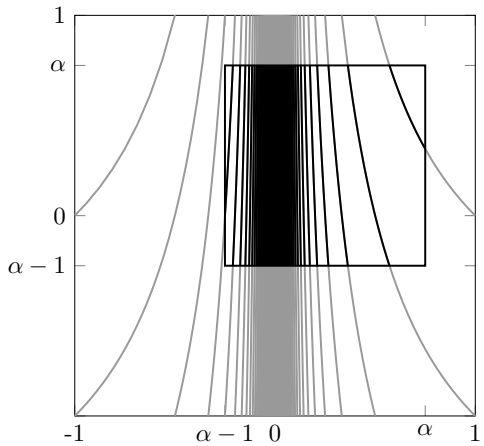


Nakada's α-CFs

For each $\alpha \in [0,1]$, define $G_\alpha : [\alpha-1, \alpha) \to [\alpha-1, \alpha)$ by $G_\alpha(0) = 0$ and

$$G_\alpha(x) = \frac{1}{|x|} - \left\lfloor \frac{1}{|x|} \right\rfloor_\alpha, \quad x \neq 0, \qquad \text{where} \quad \lfloor x \rfloor_\alpha = \lfloor x + 1 - \alpha \rfloor .$$

The α-CF expansion of $x \in [\alpha-1, \alpha)$ is $[0/1; e_1/d_1, e_2/d_2, \dots]$, where $e_{n+1} = \operatorname{sgn}(G_\alpha^n(x))$, $d_{n+1} = \lfloor 1/|G_\alpha^n(x)| \rfloor_\alpha$. Each G_α has a unique, a.c. invariant measure ρ_α, and $([\alpha-1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$ is ergodic.



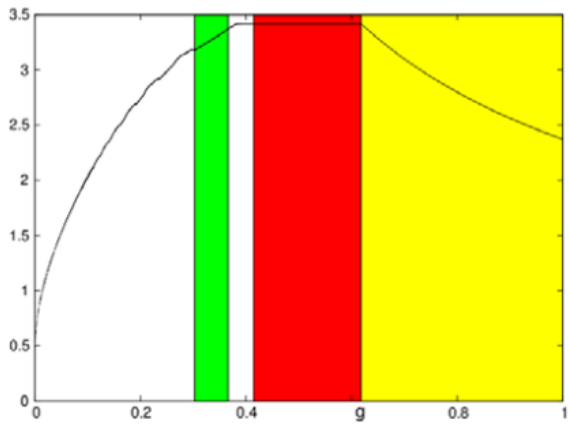
Surprising intricacies

The dynamics of G_α for $\alpha \geq 1/2$ have been well-understood since Nakada's 1981 introduction. They have been surprisingly difficult to study for $0 < \alpha < 1/2$.

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Tuning and plateaux for the entropy of α-continued fractions

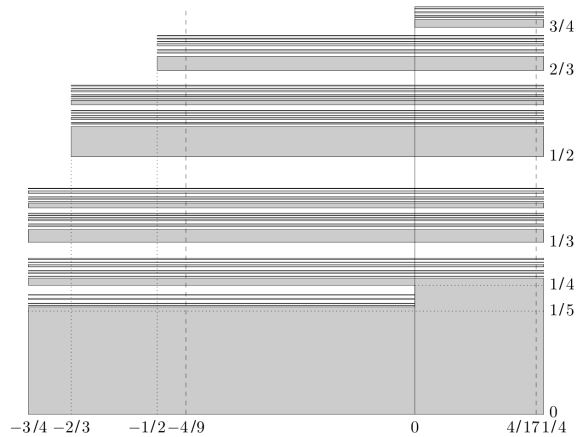
Carlo Carminati[1] and Giulio Tiozzo[2]



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Nonlinearity **25** (2012) 2207–2243 doi:10.1088/0951-7715/25/8/2207

Natural extensions and entropy of α-continued fractions

Cor Kraaikamp[1], Thomas A Schmidt[2] and Wolfgang Steiner[3]

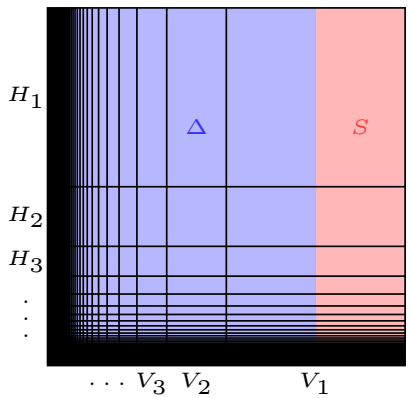


Kraaikamp's S-expansions

In 1991, Kraaikamp defined a large collection of GCF-algorithms by coupling

1. **singularisation**, which—under certain conditions—accelerates a RCF by producing a GCF whose convergents are a subsequence of the RCF-convergents, &
2. **induced transformations** of $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$, which accelerate via first-return maps.

S-expansions use induced transformations to govern singularisations of RCF-expansions: keep p_n/q_n iff $\mathcal{G}^n(x, 0) \in \Delta = \Omega \setminus S$.



Limitations of singularisation

Singularisation—which is well-known and dates back to Lagrange (1798)—is based on the identity

$$a_n + \cfrac{1}{1 + \cfrac{1}{a_{n+2} + t}} = a_n + 1 + \cfrac{-1}{a_{n+2} + 1 + t}.$$

In particular,

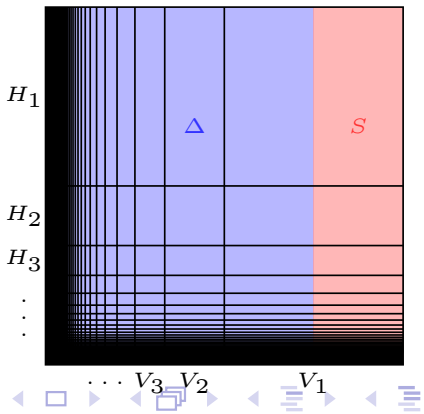
(i) can only remove p_n/q_n if $a_{n+1} = 1$, and

(ii) cannot remove p_n/q_n and p_{n+1}/q_{n+1}.

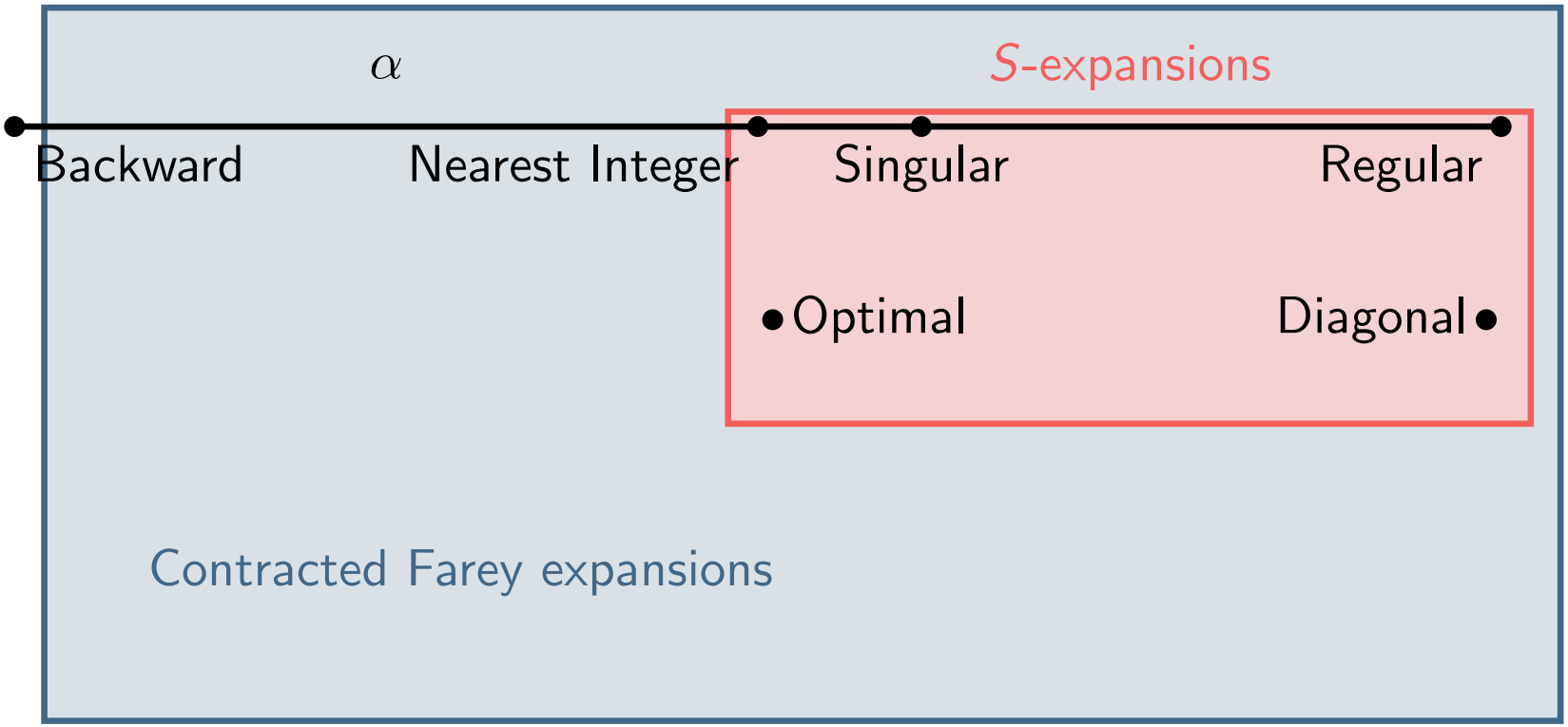
These place restrictions on Kraaikamp's **singularisation areas** S:

(i) $S \subset V_1$, and

(ii) $S \cap \mathcal{G}(S) = \varnothing$.



Unifying family



Contraction

There's a more general acceleration technique called **contraction**:

Theorem (Seidel 1855)

Let $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ *be a* GCF *with pairwise distinct convergents* P_n/Q_n. *Then for any strictly increasing sequence* $(n_k)_{k\geq 0}$ *of non-negative integers, there exists a* GCF $[\beta'_0/\alpha'_0; \alpha'_1/\beta'_1, \alpha'_2/\beta'_2, \dots]$ *whose convergents are precisely* P_{n_k}/Q_{n_k}.

Explicitly,

$$\begin{pmatrix} \alpha'_{k+1} \\ \beta'_{k+1} \end{pmatrix} = \begin{pmatrix} -\det(B_{[n_{k-1}+2,n_k+1]} Q_{[n_{k-2}+2,n_{k-1}]} Q_{[n_k+2,n_{k+1}]}) \\ Q_{[n_{k-1}+2,n_{k+1}]} \end{pmatrix}, \quad k \geq -1$$

where $B_{[m,n]} = \begin{pmatrix} P_{[m,n-1]} & P_{[m,n]} \\ Q_{[m,n-1]} & Q_{[m,n]} \end{pmatrix} := \begin{pmatrix} 0 & \alpha_m \\ 1 & \beta_m \end{pmatrix} \cdots \begin{pmatrix} 0 & \alpha_n \\ 1 & \beta_n \end{pmatrix}$, $n_k = k$ for $k < 0$ and $Q_{[-1,-2]} = 1$.

Contraction example

Let $x = \frac{\sqrt{4226}-51}{25}$. This has a periodic RCF-expansion

$$x = [0; \overline{1, 1, 3, 1, 1, 1, 4}]$$

with convergents

$$\left(\frac{p_n}{q_n}\right)_{n\geq 0} = \left(\frac{0}{1}, \frac{1}{1}, \frac{1}{2}, \frac{4}{7}, \frac{5}{9}, \frac{9}{16}, \frac{14}{25}, \frac{65}{116}, \frac{79}{141}, \frac{144}{257}, \frac{511}{912}, \frac{655}{1169}, \frac{1166}{2081}, \frac{1821}{3250}, \dots\right).$$

Let $(n_k)_{k\geq 0} = (2, 6, 9, 13, 16, 20, \dots)$. The contraction of the RCF-expansion of x w/r/t $(n_k)_{k\geq 0}$ is

$$[1/2; 6/50, -8/31, \overline{9/25, -4/31}]$$

which has convergents

$$\left(\frac{p_{n_k}}{q_{n_k}}\right)_{k\geq 0} = \left(\frac{1}{2}, \frac{14}{25}, \frac{144}{257}, \frac{1821}{3250}, \frac{18721}{33412}, \frac{236744}{422525}, \dots\right).$$

All α-CFs realised as 'S-expansions with contraction'?

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The non-monotonicity of the entropy of α-continued fraction transformations

Hitoshi Nakada[1] **and Rie Natsui**[2]

between the extensions of any α and $\alpha_{\frac{1}{2}}$ for $\alpha \in [\sqrt{2}-1, \frac{1}{2}]$ as a generalization of [1]. Here we note that the natural extension of T_α cannot be obtained by a simple induced transformation, in the sense of [1], of the natural extension of T_1. This is related to the fact that a convergent of the continued fraction expansion of x by T_α may not be a convergent of the simple continued fraction expansion of x.

Farey tent map

Define $\varepsilon : [0,1] \to \{0,1\}$ by $\varepsilon(x) = \lfloor 2x \rfloor$, and let $A_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $A_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The **Farey tent map** $F : [0,1] \to [0,1]$ is

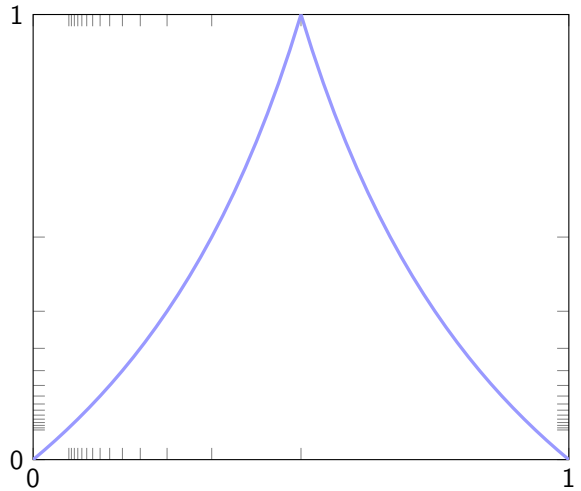
$$F(x) := A^{-1}_{\varepsilon(x)} \cdot x = \begin{cases} \frac{x}{1-x}, & \varepsilon(x) = 0, \\ \frac{1-x}{x}, & \varepsilon(x) = 1. \end{cases}$$

The dynamical system $([0,1], \mathcal{B}, \mu, F)$ is ergodic, where μ is the infinite, σ-finite, a.c. invariant measure with density $1/x$.

If $x = [0; a_1, a_2, \dots]$, then

$$F(x) = \begin{cases} [0; a_1 - 1, a_2, \dots], & a_1 > 1, \\ [0; a_2, a_3, \dots], & a_1 = 1. \end{cases}$$

In particular, $F^{a_1}(x) = G(x)$.



Farey expansions and convergents

Setting $\varepsilon_{n+1} = \varepsilon(F^n(x))$, the **Farey expansion** of x is

$$[(1-\varepsilon_1)/1; (-1)^{\varepsilon_1+1}/(2-\varepsilon_2), (-1)^{\varepsilon_2+1}/(2-\varepsilon_3), \dots].$$

Proposition

The $(n-1)^{st}$ **Farey convergent** *of* x *is* $\frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}}$, *where* j_n, $\lambda_n \in \mathbb{Z}$ *satisfy*

$$n = a_1 + \cdots + a_{j_n} + \lambda_n, \qquad 0 \le \lambda_n < a_{j_n+1}.$$

Note $F^n(x) = A_{\varepsilon_n}^{-1} \cdots A_{\varepsilon_1}^{-1} \cdot x = (A_{\varepsilon_1} \cdots A_{\varepsilon_n})^{-1} \cdot x$. One finds

$$A_{\varepsilon_1} \cdots A_{\varepsilon_n} = \begin{pmatrix} \lambda_n p_{j_n} + p_{j_n-1} & p_{j_n} \\ \lambda_n q_{j_n} + q_{j_n-1} & q_{j_n} \end{pmatrix} =: \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix},$$

so the Farey convergent u_n/s_n is read from $A_{\varepsilon_1} \cdots A_{\varepsilon_n}$.

Ito's natural extension

In 1989, S. Ito introduced an explicit natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ of $([0,1], \mathcal{B}, \mu, F)$. Here

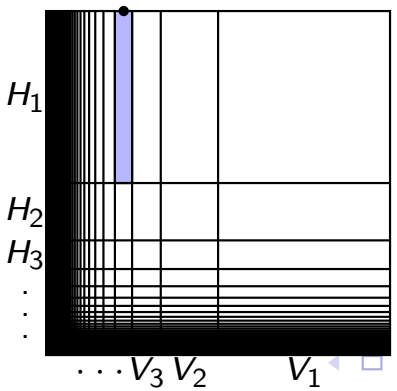
$$\mathcal{F}(z) = \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right), \qquad \bar{\mu}(A) = \iint_A \frac{dxdy}{(x+y-xy)^2}.$$

Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$



$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{F}} & \Omega \\ \pi \downarrow & & \downarrow \pi \\ [0,1] & \xrightarrow{F} & [0,1] \end{array}$$

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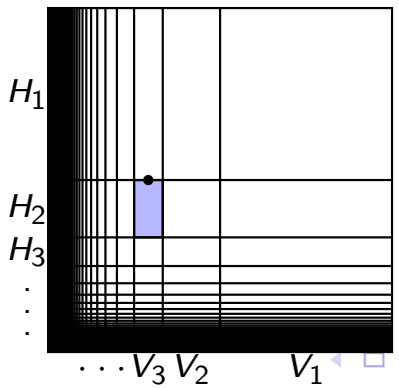
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

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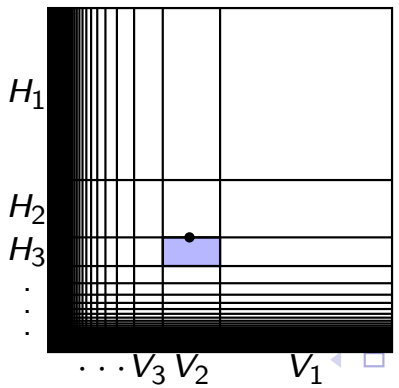
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$

$$\mathcal{F}^2(z) = ([0; 2, \overline{2, 3, 4}], [0; 3])$$



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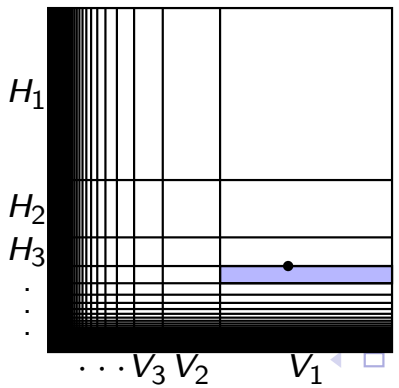
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

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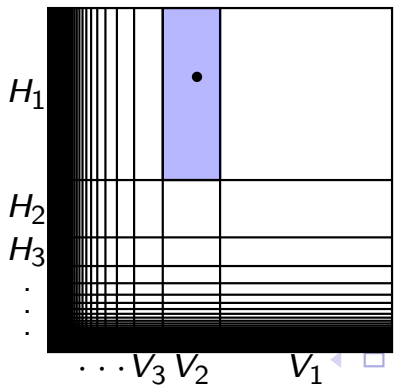
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Example

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$$\mathcal{F}(z) = \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right), \qquad \bar{\mu}(A) = \iint_A \frac{dxdy}{(x+y-xy)^2}.$$

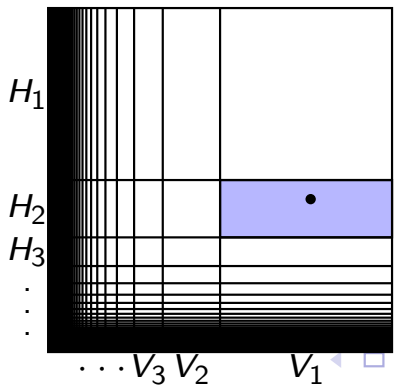
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$

$$\mathcal{F}^5(z) = ([0; 1, \overline{3, 4, 2}], [0; 2, 4])$$



$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{F}} & \Omega \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ [0,1] & \xrightarrow{F} & [0,1] \end{array}$$

Ito's natural extension

In 1989, S. Ito introduced an explicit natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ of $([0,1], \mathcal{B}, \mu, F)$. Here

$$\mathcal{F}(z) = \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right), \qquad \bar{\mu}(A) = \iint_A \frac{dxdy}{(x+y-xy)^2}.$$

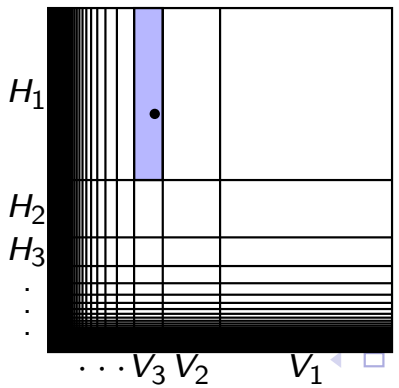
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$

$$\mathcal{F}^6(z) = ([0; 3, \overline{4, 2, 3}], [0; 1, 2, 4])$$



$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{F}} & \Omega \\ \pi \downarrow & & \downarrow \pi \\ [0,1] & \xrightarrow{F} & [0,1] \end{array}$$

Ito's natural extension

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$$\mathcal{F}(z) = \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right), \qquad \bar{\mu}(A) = \iint_A \frac{dxdy}{(x+y-xy)^2}.$$

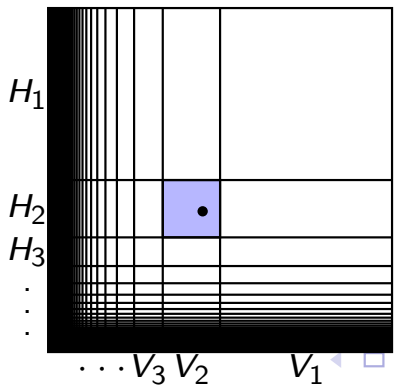
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$

$$\mathcal{F}^7(z) = ([0; 2, \overline{4, 2, 3}], [0; 2, 2, 4])$$



$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{F}} & \Omega \\ \pi \downarrow & & \downarrow \pi \\ [0,1] & \xrightarrow{F} & [0,1] \end{array}$$

Ito's natural extension

In 1989, S. Ito introduced an explicit natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ of $([0,1], \mathcal{B}, \mu, F)$. Here

$$\mathcal{F}(z) = \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right), \qquad \bar{\mu}(A) = \iint_A \frac{dxdy}{(x+y-xy)^2}.$$

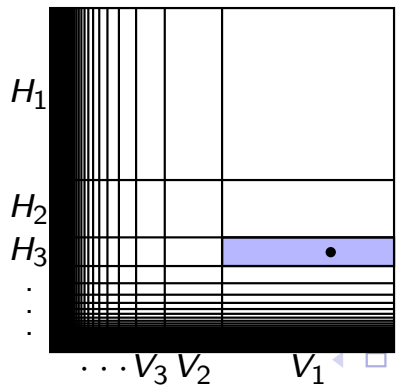
Ergodicity of F implies that of $\mathcal{F}$. Symbolically,

$$\mathcal{F}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

$$z = ([0; 4, \overline{2, 3, 4}], [0; 1])$$

$$\mathcal{F}^8(z) = ([0; 1, \overline{4, 2, 3}], [0; 3, 2, 4])$$



$$\begin{array}{ccc} \Omega & \xrightarrow{\mathcal{F}} & \Omega \\ \pi \downarrow & & \downarrow \pi \\ [0,1] & \xrightarrow{F} & [0,1] \end{array}$$

$\mathcal{F}$-orbits and Farey convergents

Letting $z = (x, 1)$, we have a 1-1 correspondence

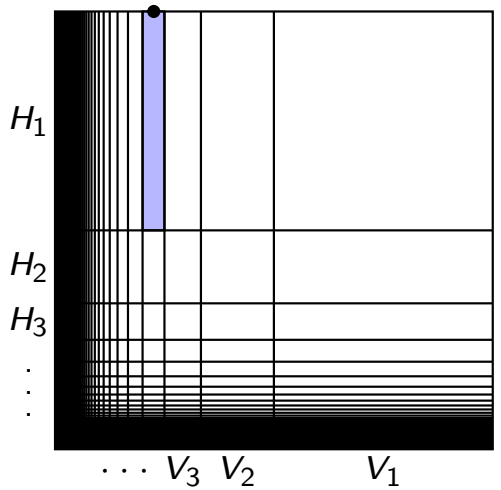
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_0 = ([0; 4, \overline{2, 3, 4}], [0; 1]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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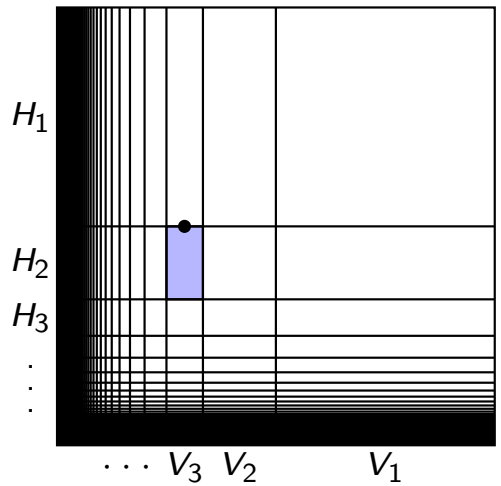
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_1 = ([0; 3, \overline{2, 3, 4}], [0; 2]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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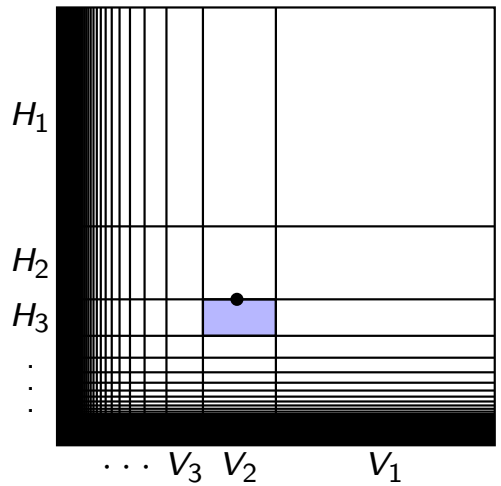
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_2 = ([0; 2, \overline{2, 3, 4}], [0; 3]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0},$$
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$\mathcal{F}$-orbits and Farey convergents

Letting $z = (x, 1)$, we have a 1-1 correspondence

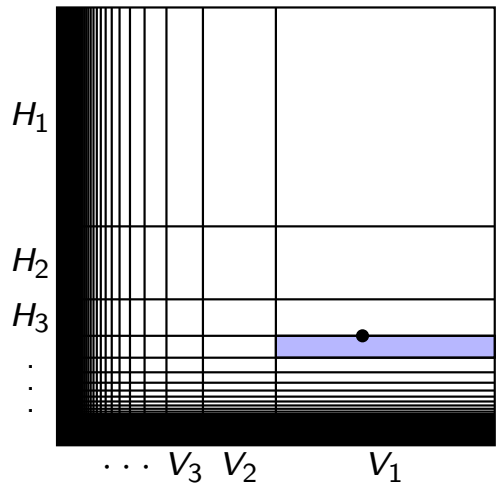
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_3 = ([0; 1, \overline{2, 3, 4}], [0; 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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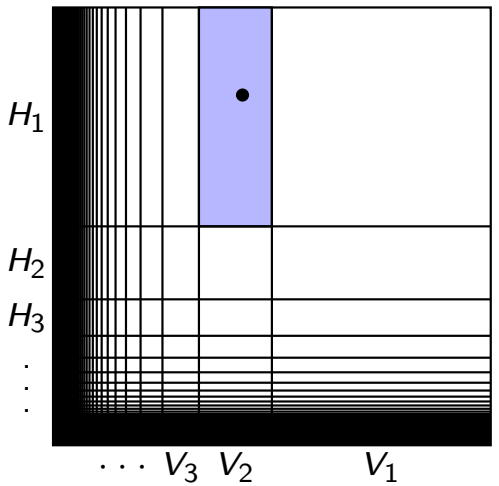
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_4 = ([0; 2, \overline{3, 4, 2}], [0; 1, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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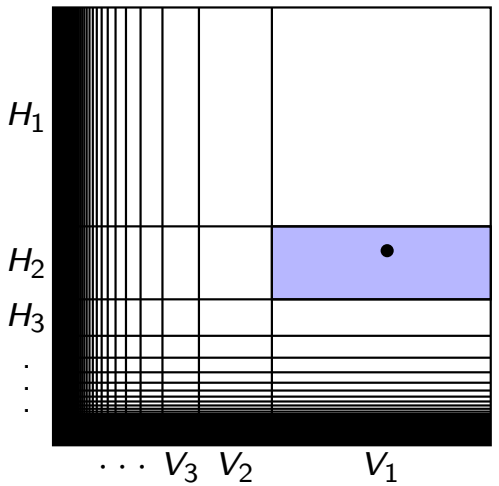
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_5 = ([0; 1, \overline{3, 4, 2}], [0; 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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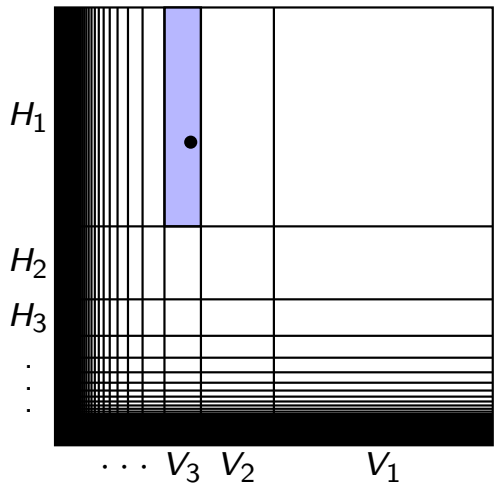
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_6 = ([0; 3, \overline{4, 2, 3}], [0; 1, 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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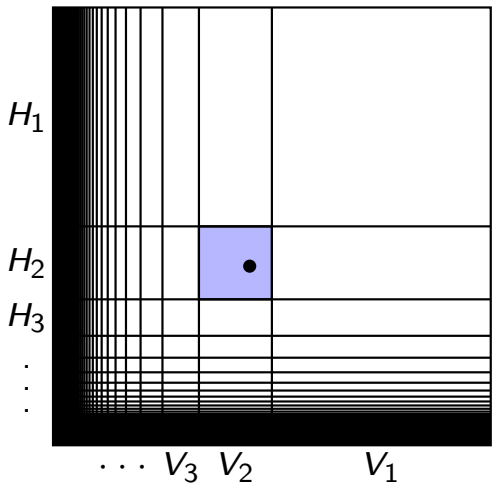
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_7 = ([0; 2, \overline{4, 2, 3}], [0; 2, 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$\mathcal{F}$-orbits and Farey convergents

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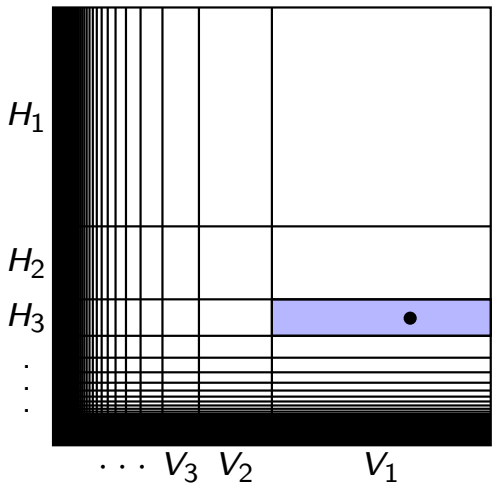
$$z_n := \mathcal{F}^n(z) \quad \longleftrightarrow \quad \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}}.$$

Example

Let $x = [0; \overline{4, 2, 3}]$. We have

$$z_8 = ([0; 1, \overline{4, 2, 3}], [0; 3, 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n \geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
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$$\left. \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \dots \right)$$



Inducing contractions

For $R \subset \Omega$ with $0 < \bar{\mu}(R) < \infty$, define $\mathcal{F}_R := \mathcal{F}^{N_R} : \Omega \to R$, where N_R is the hitting time to R. Ergodicity of $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ implies that of the **induced system** $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$, where $\bar{\mu}_R(S) := \bar{\mu}(S)/\bar{\mu}(R)$ for any measurable $S \subset R$.

Let $z = (x, 1)$. When $\bar{\mu}(\text{int}R) > 0$, $z_n = \mathcal{F}^n(z) \in R$ i.o. for Leb.–a.e. x. The map $\mathcal{F}_R$ naturally determines a subsequence $(z_k^R)_{k\geq 0}$ of $(z_n)_{n\geq 0}$ and, via $z_n \longleftrightarrow u_n/s_n$, a subsequence $(u_k^R/s_k^R)_{k\geq 0}$ of $(u_n/s_n)_{n\geq 0}$.

Definition

The **contracted Farey expansion** (CFE) of x w/r/t $R \subset \Omega$, denoted $[\beta_0^R/\alpha_0^R; \alpha_1^R/\beta_1^R, \alpha_2^R/\beta_2^R, \dots]$, is the contraction of the Farey expansion of x whose $(k-1)^{\text{st}}$ convergent is u_k^R/s_k^R, $k \geq 0$.

R determines a CF-algorithm

A priori, one must compute the Farey expansion of x to determine the CFE of x. However, one can show:

Proposition

The digits of the CFE *of* x *w/r/t* R *are* $\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z) \\ u_R(z) \end{pmatrix}$ *and for* $k \geq 0$,

$$\begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} = \begin{pmatrix} -\det A_R(\mathcal{F}_R^k(z)) s_R(\mathcal{F}_R^{k+1}(z)) \\ s_R(\mathcal{F}_R^k(z)) u_R(\mathcal{F}_R^{k+1}(z)) + r_R(\mathcal{F}_R^k(z)) s_R(\mathcal{F}_R^{k+1}(z)) \end{pmatrix}.$$

where

$$A_R(z) = \begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} = A_{\varepsilon_1} \cdots A_{\varepsilon_{N_R(z)}}$$

satisfies

$$\mathcal{F}_R(z) = (A_R(z)^{-1} \cdot x, A_1 A_R(z)^T A_1^{-1} \cdot y).$$

Regular CFs

Consider $R = H_1$, corresponding to $u_n^R/s_n^R = p_{n-1}/q_{n-1}$. One finds $A_R(z) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & a(x) \end{smallmatrix}\right)$, so $\mathcal{F}_R \cong \mathcal{G}$ via $\varphi : H_1 \to \Omega$, $(x, y) \mapsto (x, A_1^{-1} \cdot y)$.

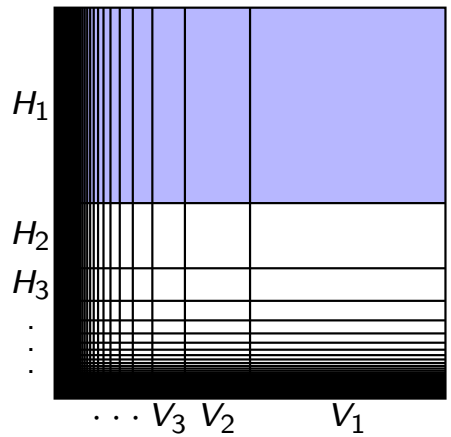
Theorem (Brown–Yin ('96))

$$(H_1, \mathcal{B}, \bar{\mu}_{H_1}, \mathcal{F}_{H_1}) \cong (\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$$

Moreover, the CFE of x w/r/t R is

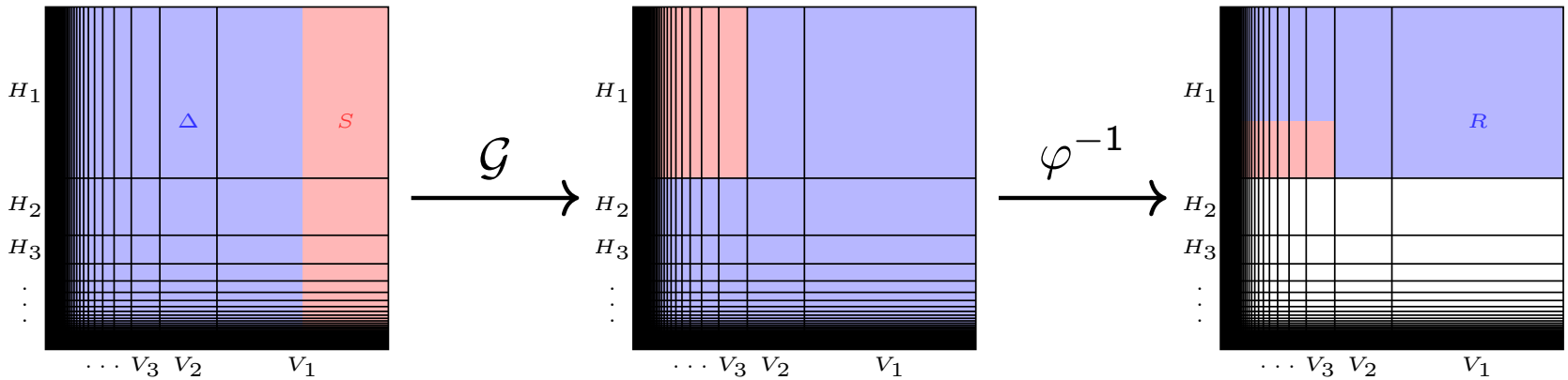
$$[0/1; 1/a(x_0^R), 1/a(x_1^R), 1/a(x_2^R), \dots],$$

where $(x_k^R, y_k^R) = \mathcal{F}_R^k(x, 1)$, i.e., we recover the RCF-expansion of x.



Kraaikamp's S-expansions

Let $S \subset \Omega$ be a singularisation area, $\Delta = \Omega \backslash S$, and recall the S-exp'n of x: keep p_n/q_n iff $\mathcal{G}^n(x, 0) \in \Delta := \Omega \backslash S$.
Setting $R := \varphi^{-1} \circ \mathcal{G}(\Delta)$ with $\varphi : H_1 \to \Omega$ as before, one finds that CFE of x w/r/t R is the precisely the S-expansion of x.



Hence also optimal, diagonal, and Nakada's α-CFs, $\alpha \geq 1/2$, are CFEs.

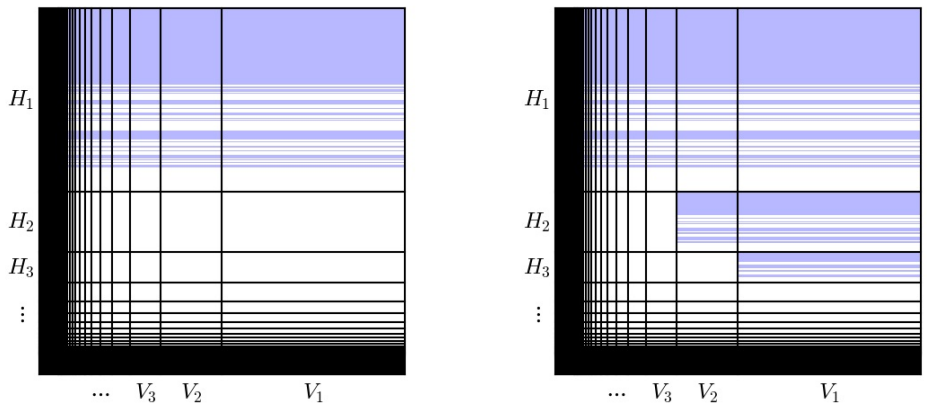
Nakada's α-CFs

Fix $0 < \alpha \leq 1$. Let $k(z) := \inf\{j > 0 \mid \mathcal{F}_{H_1}^{-j}(z) \in [0, \alpha) \times [1/2, 1]\}$, $A := \{z \in H_1 \mid \text{k(z) is odd}\}$, $A_a := A \cap ([\alpha, 1] \times [0, 1]) \cap V_a$, and

$$R := A \cup \bigcup_{a=2}^{\infty} \bigcup_{\lambda=1}^{a-1} \mathcal{F}^{\lambda}(A_a).$$

Theorem (Dajani, Kraaikamp, S. 2025)

$(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is the natural extension of $([\alpha - 1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$. Moreover, the CFE *of x w/r/t R is the α-*CF *expansion of x.*



Nakada's α-CFs, cont'd

Idea.

Let $\sigma : [\alpha - 1, \alpha) \to [0, 1)$ where $x \mapsto x \pmod 1$. One finds for $x = [0; a_1, a_2, \dots]$ that

$$\sigma \circ G_\alpha \circ \sigma^{-1}(x) = \begin{cases} F^{a_1}(x), & x < \alpha, \\ F(x), & \alpha \le x \le 1/2, \\ F^{a_1+a_2}(x), & \max\{\alpha, 1/2\} < x < 1. \end{cases}$$

The region R is constructed so that

$$\mathcal{F}_R(z) = \begin{cases} \mathcal{F}^{a_1}(z), & x < \alpha, \\ \mathcal{F}(z), & \alpha \le x \le 1/2, \\ \mathcal{F}^{a_1+a_2}(z), & \max\{\alpha, 1/2\} < x < 1. \end{cases}$$

Hence if $\pi(x, y) = x$, we have $(\sigma^{-1} \circ \pi) \circ \mathcal{F}_R = G_\alpha \circ (\sigma^{-1} \circ \pi)$. □

Applications

1. One can define new, **superoptimal** CFs whose convergents u_n^R/s_n^R satisfy specified Diophantine properties ($s_n^R|s_n^R x - u_n^R| < \varepsilon$) and speed of convergence ($\lim \frac{1}{n} \log s_n^R > C$ a.s.) (S. ('26)).
2. One can show the metric entropy satisfies $h(\mathcal{F}_R)\bar{\mu}(R) = \pi^2/6$, simultaneously extending results for S-exp'ns (Kraaikamp ('91)) and α-CFs (Kraaikamp-Schmidt-Steiner ('12)).
3. Using the above, one can (presumably) show continuity of $h(G_\alpha)$ as a function of α (Kraaikamp-Schmidt-Steiner ('12), Tiozzo ('14)).
4. As a corollary of results of Heersink ('19), each G_α is the factor of the first-return of the geodesic flow to certain cross-section of $\mathrm{T}^1(\mathcal{M})$ (Arnoux-Schmidt ('13)).

Questions

1. Can this construction of the nat'l ext'n of G_α be used to prove the set of G_α-normal numbers does not depend on α (Dajani-Kraaikamp-Nakada-Natsui ('25))?
2. Are the nat'l ext'ns of Tanaka-Ito α-CFs, Katok-Ugarcovici (a, b)-CFs ($b = a - 1$), α-Rosen CFs realised via similar constructions?

Merci